

1. Compute $\int_0^1 x^{29} \ln x \, dx$.

Answer: $-\frac{1}{900}$

Solution: Integration by parts gives that $\int x^{29} \ln x \, dx = \frac{x^{30} (30 \ln x - 1)}{900}$ - evaluating gives us

$$\boxed{-\frac{1}{900}}.$$

2. Compute $\int_{-\pi}^{\pi} |x - \sin x| \, dx$.

Answer: $\pi^2 - 4$

Solution: $\int_{-\pi}^{\pi} |x - \sin x| \, dx = 2 \int_0^{\pi} (x - \sin x) \, dx = \boxed{\pi^2 - 4}.$

3. Compute:

$$\sum_{k=0}^{\infty} \frac{(k+1)^3}{k!}.$$

Answer: $15e$

Solution: It is obvious that this sum should be some version of the exponential function. It can be resolved as derivatives of the exponential by noticing:

$$(k+1)^3 = 1 + 7k + 6k(k-1) + k(k-1)(k-2)$$

Inserting this we can see that these represent 0 to 3 derivatives of the exponential function evaluated at 1. The result follows.